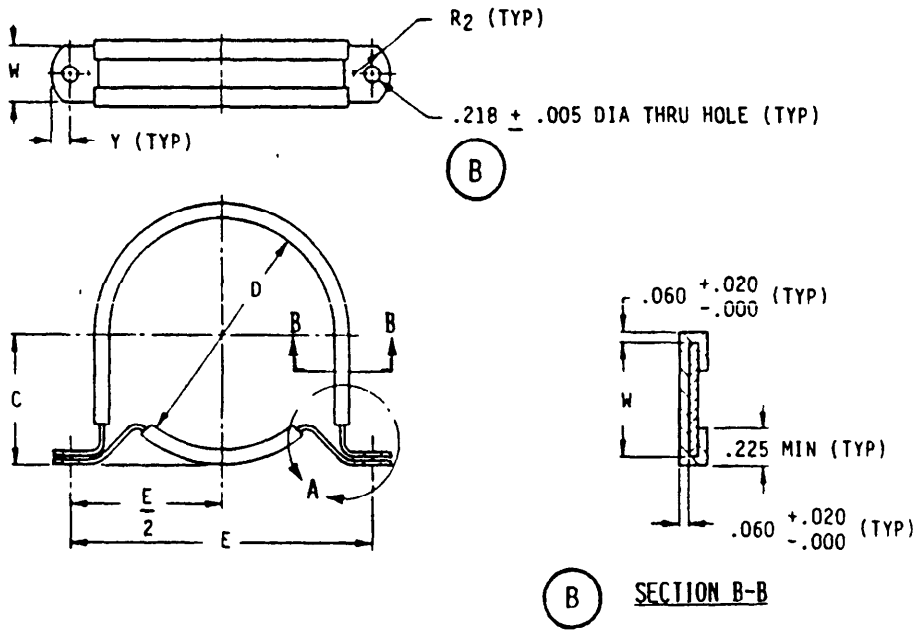
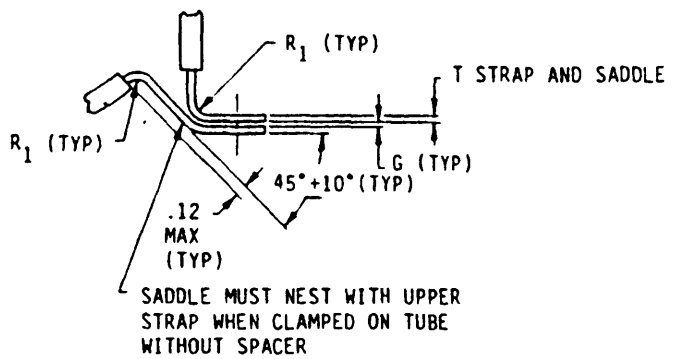
MILITARY SPECIFICATION SHEET

CLAMP ASSEMBLY, SADDLE-TYPE, CUSHIONED, NBR RUBBER, 321 CRES, 275 F, FUEL RESISTANT, GENERAL PURPOSE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the clamps described herein shall consist of this document and the latest issue in effect of specification MIL-C-85449.





TOLERANCES: .XXX ± .010
.XX ± .03
UNLESS OTHERWISE SPECIFIED

DETAIL A

AMSC N/A

FSC 5340

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TABLE I. CLAMP DIMENSIONS

DASH NO.	TUBE SIZE	C ± .020	D [5]	E ±.015 [5]	G ±.001 [7]	R_1	R_2	T [6]	W +.015 -.000	Y	WEIGHT, LBS. MAX.
-10	5/8	0.493	.625	1.525	.020	.090	.37	.040	.500	.218	0.0437
-12	3/4	0.555	.750	1.650							0.0483
-14	7/8	0.618	.875	1.775							0.0530
-16	1	0.690	1.000	1.958	.030	.109	.50	.050	.625	.250	0.0847
-18	1 1/8	0.753	1.125	2.083							0.0912
-20	1 1/4	0.815	1.250	2.208							0.0977
-24	1 1/2	0.940	1.500	2.458							0.1106
-28	1 3/4	1.065	1.750	2.708							0.1235
-32	2	1.203	2.000	3.016		.125		.063	.750		0.1941
-36	2 1/4	1.328	2.250	3.266							0.2120
-40	2 1/2	1.453	2.500	3.516							0.2300
-44	2 3/4	1.578	2.750	3.766							0.2479
-48	3	1.703	3.000	4.016							0.2658

(B)

REQUIREMENTS:

(B) Materials: Strap and Base - 321 CRES sheet per AMS 5510, except that a tensile strength of 100-120 KSI shall be required.

Cushion - Nitrile butadiene rubber, 65-75 durometer, color-yellow, per the general specification plus the requirements listed in Table II.

(B) Finish: Strap and base, ball burnish to round edges to a .010 radius, and passivated per QQ-P-35.

Marking: Impression stamp complete part number and manufacturer's trademark on strap, visible with cushion in place above the center line. The saddle shall have the dash size impression stamped in the vicinity of the mounting hole away from any bend radii.

Part No: Part numbers shall consist of the following: (In sequence)

1. The letter "M"
2. The basic specification number
3. A slash and slash number of this specification sheet
4. A dash and the appropriate size dash numbers from Table I

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Example:

(B) M85449 /1 - 16

- 16 — dash number (-16 = 1" Tube Size)
- /1 — slash sheet number
- M85449 — basic military specification number

CUSHION REQUIREMENTS:

1. Physical properties - as specified in Table II. Unless otherwise specified, results are an average of 5 specimens and tolerance of temperature is ± 5°F.

TABLE II. CUSHION MATERIAL PHYSICAL PROPERTIES

TEST	TEST METHOD	REQUIRED ORIGINAL PROPERTIES	ALLOWABLE CHANGE FROM ACTUAL ORIGINAL PROPERTIES AFTER			
			* FUEL IMMERSION JP-4 168 HRS AT +100°F	* HYDRAULIC FLUID IMMERSION MIL-H-5606 70 HRS. AT +158°F	* HYDRAULIC FLUID IMMERSION MIL-H-83282 70 HRS. AT +158°F	HEATING AGING 70 HRS. AT +275°F
HARDNESS, DUROMETER "A"	ASTM D2240	65 - 75	-20 MAX.	-15 MAX.		+10% MAX.
TENSILE STRENGTH - PSI	ASTM D412	2000 MIN.	-30% MAX.	-10% MAX.		-40% MAX.
ELONGATION - %	ASTM D412	500 MIN.	-10% MAX.	-10% MAX.		-75% MAX.
TEAR STRENGTH - PPI	ASTM D624 DIE "B"	300 MIN.	-35% MAX.	-10% MAX.		-40% MAX.
SPECIFIC GRAVITY	----------	AS MEASURED	---------	---------		±2%
VOLUME CHANGE - %	ASTM D471-75					
1. AFTER CONDITIONING		----------	+15% MAX.	+10% MAX.		-10% MAX.
2. AFTER 24 HRS. AIR DRY		----------	+10% MAX.	+10% MAX.		---------

* Property measurements shall be performed immediately after removal from the immersion bath.

2. Ozone Resistance - See MIL-C-85449

3. Compression Set - See MIL-C-85449
   - Air age at 212°F
   - Not to exceed 55% average of 3 specimens

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4. Flammability: - See MIL-C-85449

- Specimens: -16 size clamp assemblies unbent until straight

- Vertical burn test only

NOTES:

1. Dimensions in inches unless otherwise specified.

2. Metal parts shall be free of burrs, sharp edges, tool marks and die marks.

3. With clamp assembled on tube, no gap is permitted between cushion and tube for upper half of strap and cushioned length of saddle.

4. With cushion pushed to either end of strap, sufficient cushion shall be available to prevent strap contacting tube.

5 The "D" & "E" dimensions will be measured with mandrel clamped and a "G" thickness spacer between the clamp feet.

B 6 Tolerances per AMS 2242.

7 "G" dimension is for thickness of shim required to inspect assembled clamp.

B 8. In the event of a conflict between the text of this specification sheet and the references cited herein, the text of this specification sheet shall take precedence.

INTENDED USE:

These clamps are intended for use as follows:

Temperature Range: -65°F to +275°F (For higher temperature rated clamps, see MIL-C-85449/3 Clamps).

Systems:

1. All aromatic fuel system applications requiring a high degree of cushion exposure to system fluid.

2. All fluid and electrical systems on aircraft and aerospace vehicles using petroleum based hydraulic systems.

3. In electrical systems for support of wire bundles provided multiple wire bundles are wrapped or contained in conduits or other wire protection tubing.

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4. In systems with high vibration areas such as near pumps, motors, etc. It is to be noted that MIL-C-85449 requires vibration tests only up to the -24 size clamp. Larger clamps used in high vibration areas should be subjected to additional testing.

Custodians:
Navy - AS
Air Force - 99
Army - AV

Reviewer Activities:
Air Force - 82
DLA - IS

Preparing Activity:
Navy - AS

Project No.: 5340-1831

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